7.2.R1 (1/1 point)

Why are natural cubic splines typically preferred over polynomials of degree d?

- [ ] Polynomials have too many degrees of freedom
- [x] Polynomials tend to extrapolate very badly ✔
- [ ] Polynomials are not as continuous as splines

EXPLANATION

Polynomials can oscillate wildly once they get outside

the boundaries of the data set. Natural splines, on the other hand,

always extrapolate linearly.

Hide Answer *You have used 2 of 5 submissions*

7.2.R2 (1 point possible)

Let $1\{x \le t\}$ denote a function which is 1 if $x \le t$ and 0 otherwise.

Which of the following is a basis for linear splines with a knot at t? Select all that apply:

- [x] $1, x, (x-t)1\{x > t\}$ ✔
- [ ] $1, x, (x-t)1\{x \le t\}$ ✔
- [ ] $1\{x > t\}, 1\{x \le t\}, (x-t)1\{x > t\}$
- [ ] $1, (x-t)1\{x \le t\}, (x-t)1\{x > t\}$ ✔

EXPLANATION

Every function in the basis must be continuous at t, and

we must be able to represent any piecewise linear function as a linear combination of the basis.

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